



Bahir Dar University
Faculty of Social Science
Department of Geography and Environmental studies
Masters of Science (MSc.) In Geo-Information Science

Fundamentals of Remote Sensing (Geis633)

Project Title: Object-Based Image classification (OBIC) for Mapping Land Fragmentation and Crop Field Delineation, Bata kebele, Bahir Dar Area

Conducted By: Mekonnen Gidey (ID: BDU1806929)

Awoke Amsalu (ID: BDU1806924)

Submitted to: Dr. Daniel Ayalew

Date: February 13, 2026

❑ Introduction

- Agriculture is the backbone of Ethiopia's economy, supporting livelihoods for over 70% of the population and contributing significantly to GDP. However, **land fragmentation** the division of farmland into small, scattered plots has become a major constraint to productivity.
- **Agricultural land fragmentation** refers to the division of farmland into smaller and spatially dispersed parcels.
- Remote sensing and GIS technologies have been widely used to monitor land use and agricultural systems.
- However, **pixel-based classification** methods classify individual pixels independently, often producing noisy outputs in heterogeneous smallholder landscapes.

❑ Object-Based Image classification (OBIC)

- It represents an advanced alternative, grouping pixels into meaningful image objects prior to classification.
- OBIA integrates **spectral reflectance, geometry, texture, and contextual relationships**, making it highly suitable for fragmented agricultural landscapes.
- **Think objects, not pixels.**
- Traditional **pixel-based** image_classification assigns a land cover class per pixel. All pixels are the same size, same shape and don't have any concept of their neighbors.
- However, OBIA segments an image grouping small pixels together into vector objects.
- **Segmentation:** Break the image up into objects representing land-based features.
- **Classification:** Classify those objects using their shape, size, spatial and spectral properties.

❑ Objectives

- **General Objective**

- The primary objective of this project is to build Object-Based Image Classification (OBIC) model within ERDAS IMAGINE OBJECTIVE, which can accurately delineate agricultural plot boundaries and provide statistical data for fragmentation analysis.

□ Study Area

- The study was conducted in the Bahir Dar area, located in the Amhara National Regional State of northwestern Ethiopia.
- The area is characterized by smallholder mixed farming systems dominated by cereals such as teff, maize, and wheat etc.
- Agricultural parcels are typically small and irregular, making it an ideal site for OBIA-based fragmentation analysis.

❑ **Technical Methodology and Process**

- The development of Object-Based Image classification (OBIC) was implemented using

ERDAS IMAGINE 2015 – IMAGINE Objective.

- With the following a structured workflow consisting of:

1. Raster preprocessing,
2. Object segmentation,
3. Feature extraction,
4. Classification,
5. Raster-to-vector conversion,
6. Geometric processing,
7. Fragmentation metric computation

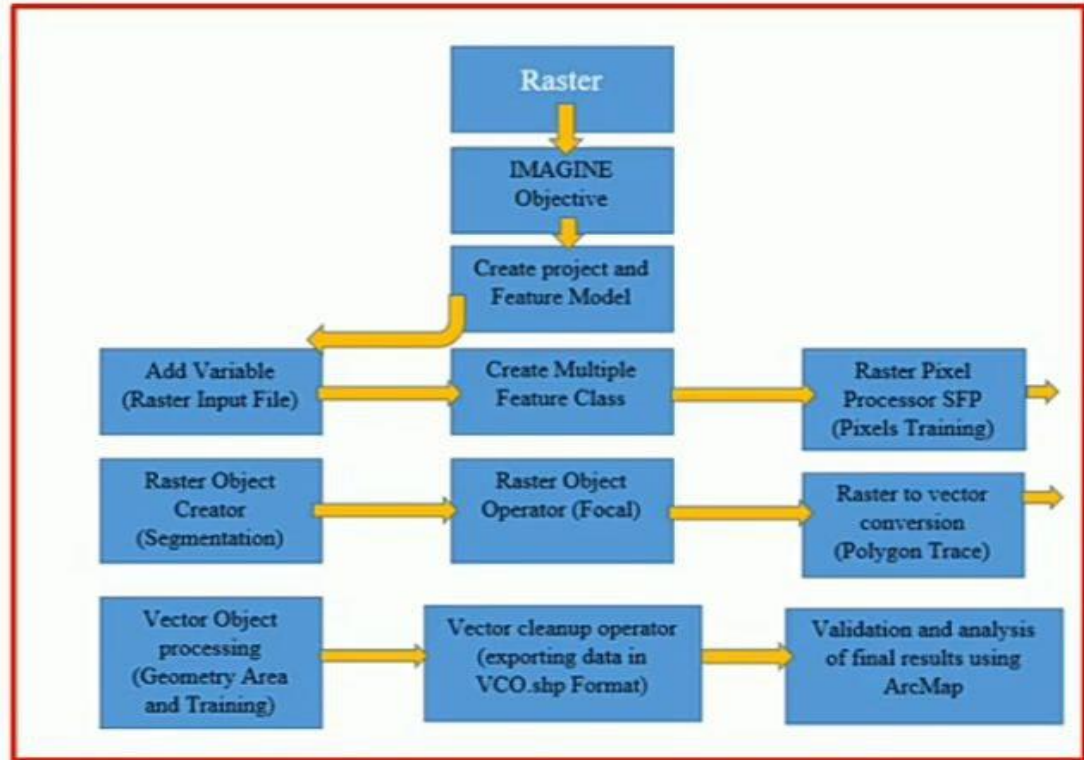
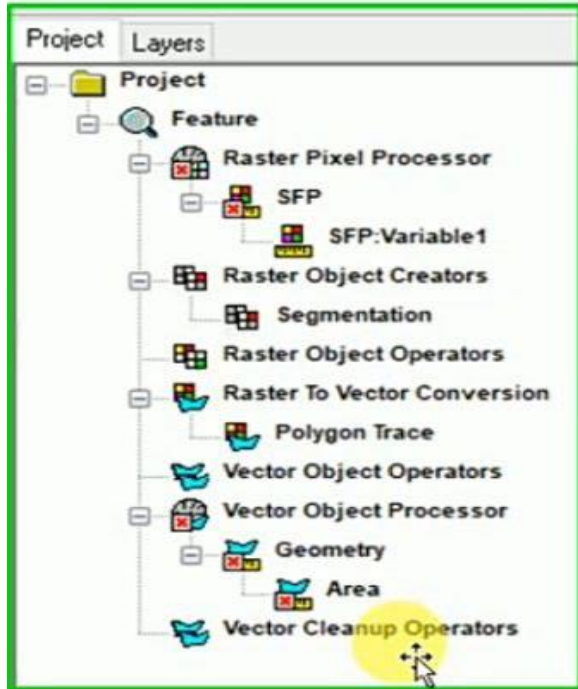


Figure 2: Object-Based Classification Workflow in ERDAS IMAGINE 2015

Variable Creation (Feature Variables Setup) And Training

feature project.lfp - feature model.lfp - Objective Workstation

File View Tools Process Help

Project Layers

- Project
 - Feature
 - Raster Pixel Processor
 - Raster Object Creators
 - Raster Object Operators
 - Raster To Vector Conversion
 - Vector Object Operators
 - Vector Object Processor
 - Vector Cleanup Operators

Properties Training I/O

Pixel Classifier: Multi Bayesian Network

Input Raster Variables: Variable1

Available Pixel Cues:

SFP

Add the selected Available Pixel Cue to the Current Pixel Cue Metrics

Row	Value	Histogram
0	5	2
1	6	1
2	7	3
3	8	5
4	9	42
5	10	34
6	11	7
7	12	78

Layers

Project

- Feature
 - Raster Pixel Processor
 - SFP
 - SFP-Variable1
 - Raster Object Creators
 - Raster Object Operators
 - Raster To Vector Conversion
 - Vector Object Operators
 - Vector Object Processor
 - Vector Cleanup Operators

Classes

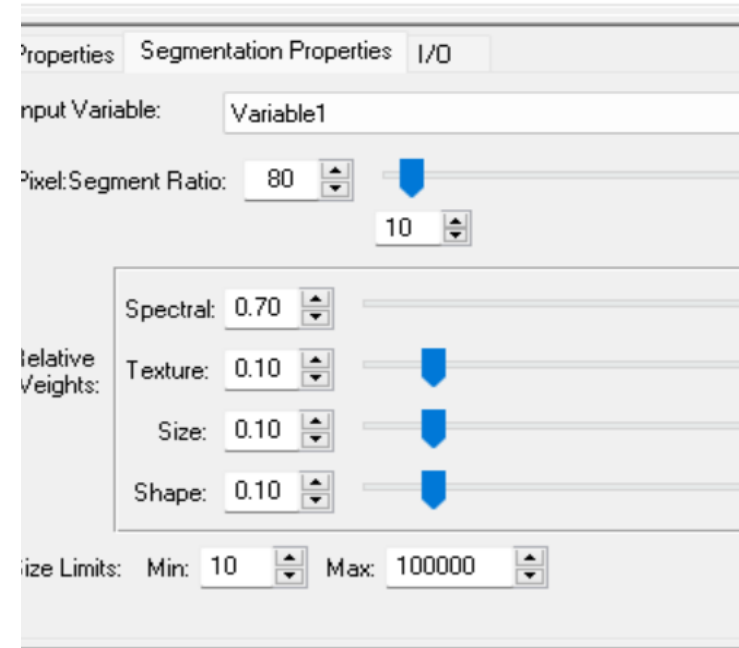
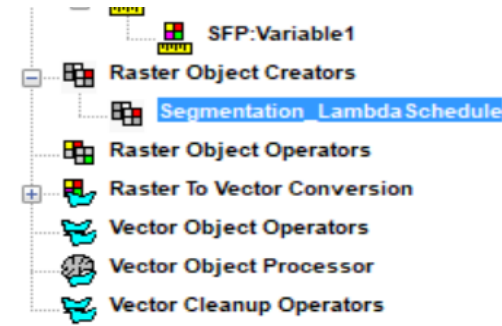
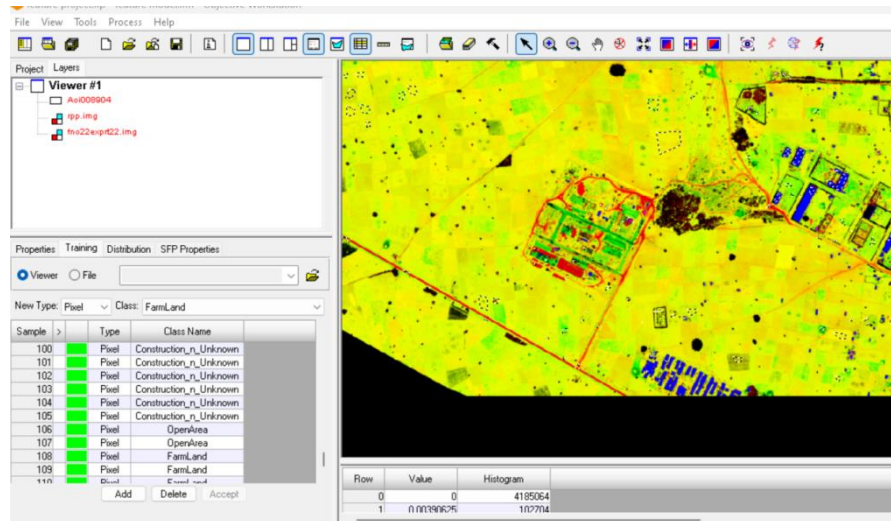
Name	Color
parcel	
roof	
tree	
Road	
Vehicle	
OpenLand	

Row Histogram

Row	Histogram
0	0
1	0
2	0
3	0
4	0
5	0
6	0
7	0

3. Raster Object Creation (Segmentation Using FLS)

- The **Full Lambda Schedule (FLS)** segmentation algorithm was applied to generate homogeneous image objects.
- The output consisted of spatially coherent raster objects representing potential agricultural parcels.



4. Raster Object Operators (Feature Extraction)

- After segmentation, object-level attributes were extracted using Raster Object Operators.
- These features formed the basis for object classification.

5. Creation of Feature Classes and Object-Based Classification

- Using the Feature Manager: The following land cover classes were defined:
 - ✓ Crop fields
 - ✓ Bare land
 - ✓ Built-up area
 - ✓ Natural vegetation

6. Polygon Trace (Raster to Vector Conversion)

- Raster objects were converted into vector polygons using the **Polygon Trace** operator.

This process:

- ✓ Converted classified raster segments to polygon shapefiles.
- ✓ Preserved parcel boundaries.
- ✓ Enabled geometric analysis.

7 vector object processor

The screenshot displays the 'Vector Object Processor' software interface. The top section shows a workflow tree with the following components:

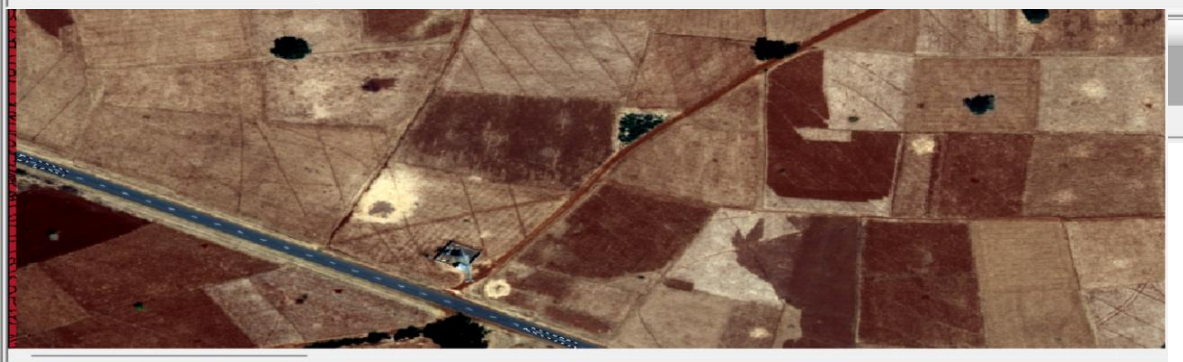
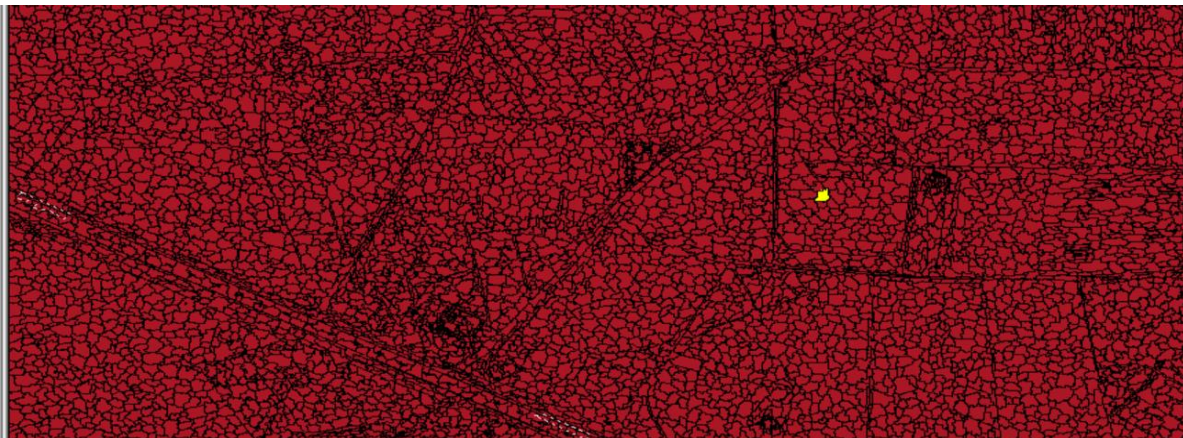
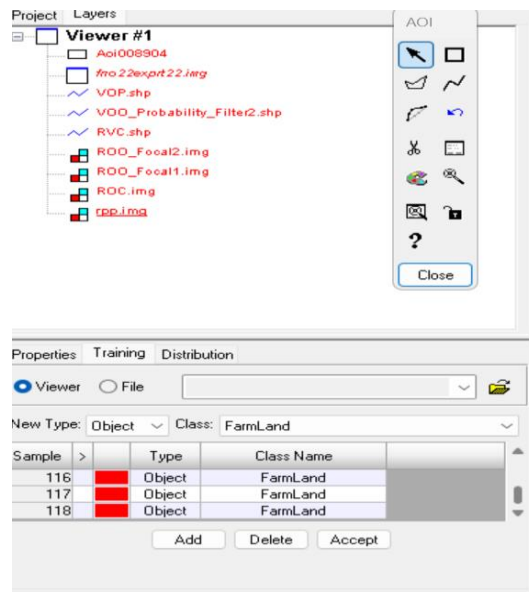
- Vector Object Operators
 - Probability Filter
- Vector Object Processor
 - Geometry
 - Area
 - Axis2/Axis1
 - Rectangularity
 - Compactness
- Vector Cleanup Operators

The bottom section contains configuration settings:

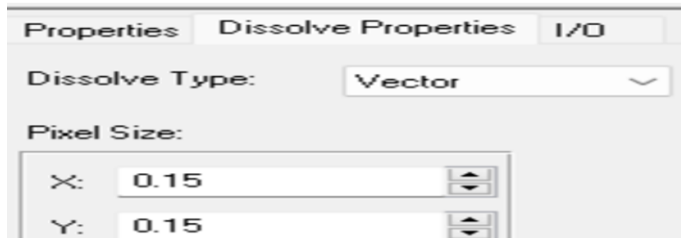
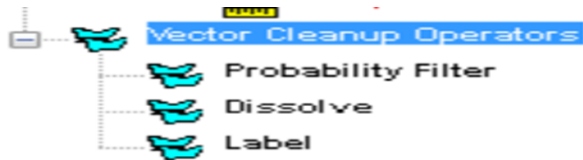
- Properties** | **Training** | **Distribution**
- Object Classifier: Multi Bayesian Network
- Input Variable: (empty)
- Available Object Cues: Geometry:Rectangularity
- Pixel Probability Weight: 50 (slider from 0 to 100)
- ☐ Enforce Distribution Bounds

A red map is visible on the right side of the interface.

...vector object processing



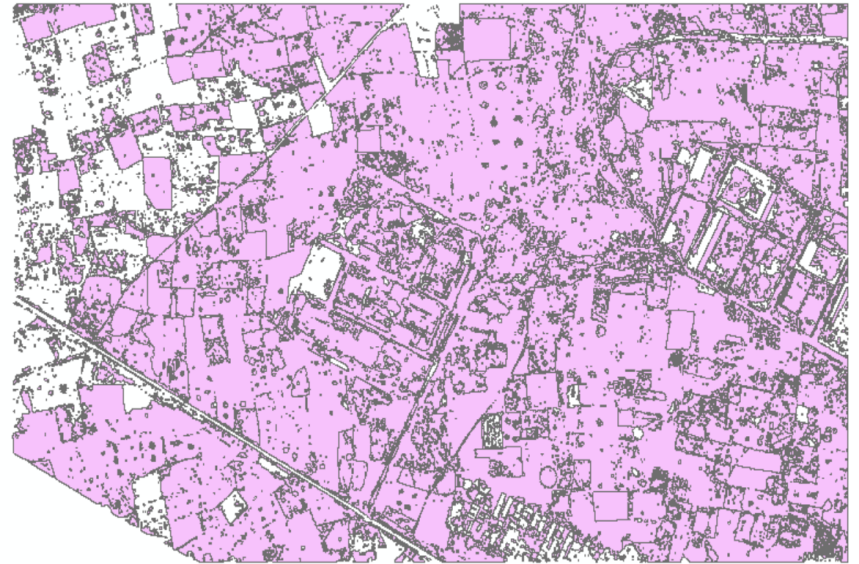
8 vector cleanup operators



...Vector Cleanup Operation result

- Final cleaning operations included: **Deleting artifacts, Simplifying geometry, Ensuring topological correctness**
- The resulting dataset represented the final crop field delineation layer.

A	B	C	D	E	F	G	H	I	J
OID	FID_1	PixProb_ Fa	PixProb_ Ro	PixProb_ Op	PixProb_ _1	PixProb_ Tr	PixProb_ Co	Area	Axis2_Axi s
-1	1	0.9921	0	0.9749	0.7905	0.0877	0.9851	91.2891	0.5047
-1	2	0.9851	0	0.9389	0.8517	0.1915	0.9814	18.1222	0.7192
-1	3	0.9926	0	0.8575	0.4528	0	0.9428	2.7906	0.6348
-1	4	0.9877	0	0.857	0.4328	0.0807	0.9241	16.364	0.6194
-1	5	0.9947	0	0.9736	0.8199	0.1343	0.9834	222.6119	0.9239
-1	6	0.9843	0	0.941	0.8903	0.226	0.9737	141.2558	0.7153
-1	7	0.9975	0	0.9808	0.8771	0.1751	0.9941	39.8172	0.6714
-1	8	0.9932	0	0.9585	0.9108	0.2864	0.9848	68.5281	0.5368
-1	9	0.9898	0	0.9756	0.7172	0.099	0.9857	84.4953	0.3344
-1	10	0.9898	0	0.9411	0.9207	0.329	0.9691	134.4259	0.2542
-1	11	0.9953	0	0.9603	0.9163	0.33	0.9496	7.812	0.475
-1	12	0.9945	0	0.969	0.9803	0.6931	0.864	7.3619	0.84



RESULTS AND ACCURACY

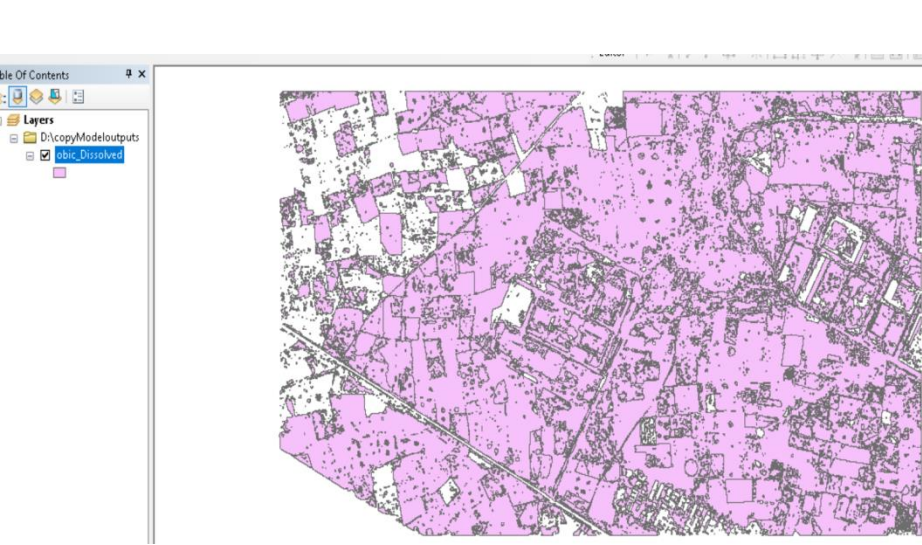
Statistical Validation

Based on the internal metrics of the ERDAS Objective Bayesian Network:

Class	Mean PixProb (Spectral)	Mean ObjProb (Geometry)	Confidence
FarmLand	0.992	0.561	High
Road	0.955	0.589	Medium
Construction	0.904	0.647	Medium

high **PixProb (0.99)** for farmland indicates that the spectral training was nearly perfect at **distinguishing farmland from other covers.**

The lower **ObjProb** reflects the geometric complexity of **traditional Ethiopian plots, which do not always follow perfect rectangles**



Land Fragmentation Analysis

The model identified **7,420 individual plots**. The **Area** attribute in the table reveals high variance, with many sub-plots falling significantly below the 0.84 ha benchmark, confirming the presence of extreme fragmentation in the study area.

❑ Results and Discussion

DISCUSSION: CHALLENGES AND LIMITATIONS

Despite the good result of the model, the "perfect" extraction and delineation of smallholder plots remains challenging.

- **Over-Segmentation:** Because 0.15m imagery is so detailed, the model sometimes identified a single field as multiple small polygons because of a plow line or a small tree in the center. Our CSV shows objects as small as 2.7 m^2 , which are clearly sub-field fragments. **Spectral Confusion:** In the vector image, we observed that dirt roads and fallow fields have similar spectral values. This resulted in some roads being classified as farmland.
- **Boundary Generalization:** The Vector Dissolve stage (Step 10) is critical. If the dissolve threshold is too low, the map remains "noisy." If it is too high, small valid fields are merged into their neighbors.

Conclusion

This study demonstrated the effectiveness of Object-Based Image Analysis (OBIA) combined with the Full Lambda Schedule (FLS) segmentation algorithm for automated crop field delineation in the Bahir Dar area.

From Applications perspective: such model plays key role for developing downstream applications. For example, yield prediction, Land Consolidation decision: Identifying regions where plots are too small for mechanization. Crop mapping and Yield prediction. Cadastral Updating: Automating the update of land tenure maps in the Amhara region. Policy Support: Providing spatial evidence for land administration reforms.

References

- Blaschke, T., Lang, S., & Hay, G. J. (Eds.). (2008). *Object-Based Image Analysis: Spatial Concepts for Knowledge-Driven Remote Sensing Applications*. Springer.
- Hay, G. J., et al. (2009). Object based image analysis for remote sensing. *ISPRS Journal of Photogrammetry and Remote Sensing*, 65(1), 2–16.
- Li, Q., Wang, C., Zhang, B., & Lu, L. (2015). Object-based crop classification with Landsat-MODIS enhanced time-series data. *Remote Sensing*, 7(12), 16091–16107.
- Tetteh, G. O., Schwieder, M., Erasmi, S., Conrad, C., & Gocht, A. (2023). Comparison of OBIA and deep learning for agricultural field delineation. *PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science*.
- Zhou, Y., et al. (2018). A comparison of OBIA approaches for field boundary delineation. *Computers and Electronics in Agriculture*.
- Blaschke (2008) and Hay et al. (2009) *Object-based Image Analysis*
- Tetteh et al. (2023) and Zhou et al. (2018) *comparisons of segmentation and delineation methods*.